



AI-Powered Retail Sales Forecasting & Analysis

A data-driven approach to predicting near-term sales by store and category — using 2 years of daily transaction data, SQL exploration, and machine learning to inform inventory and marketing decisions.

21,930 RECORDS

5 STORES

6 CATEGORIES

\$38.4M REVENUE

The Business Problem

Retailers lose money in two ways when they can't predict demand. The guiding question: **can we forecast near-term sales accurately enough, by store and category, to actually inform inventory and marketing decisions?**

Overstocking

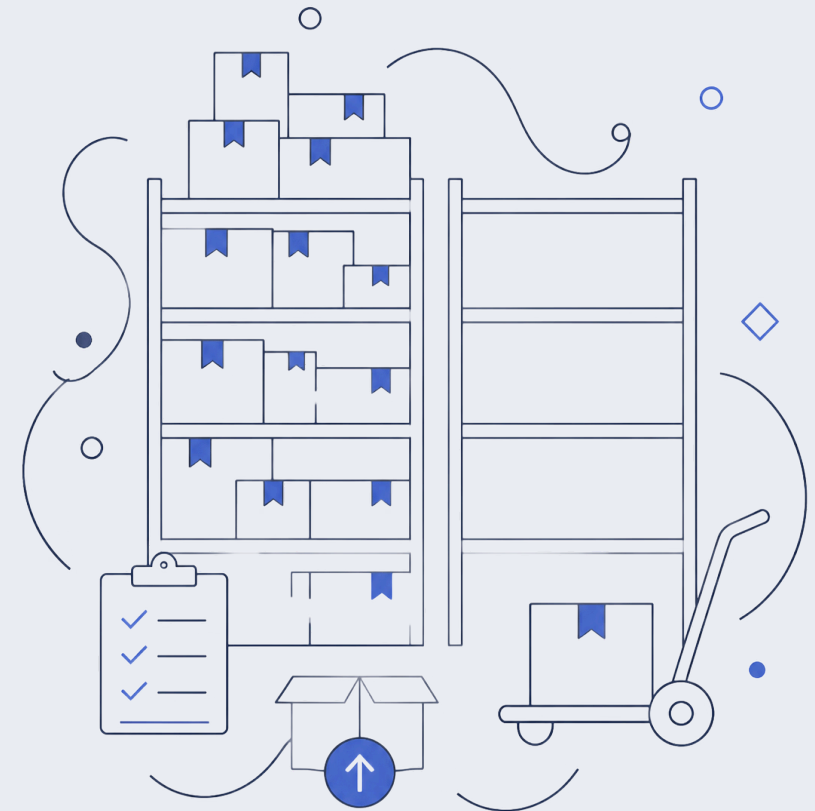
Ties up cash, forces markdowns, and creates waste — capital locked in unsold inventory.

Understocking

Lost sales, especially during peak periods when demand spikes unexpectedly.

Wasted Marketing Spend

Campaigns not timed to when demand is actually rising deliver poor returns.



How I Solved It

A structured six-step pipeline from raw data to a production-ready forecast.

01

Load Data into SQL

2 years of daily sales (5 stores, 6 categories) loaded into a proper SQLite database.

03

Engineer Features

Lag values, rolling averages, and calendar flags for holidays and promotions.

05

Time-Respecting Evaluation

Tested on Nov–Dec 2024 (Black Friday + Christmas) — not randomly shuffled.

02

Explore with SQL First

Trends, weekday patterns, category/store performance, and promo effects — before any modeling.

04

Test Multiple Models

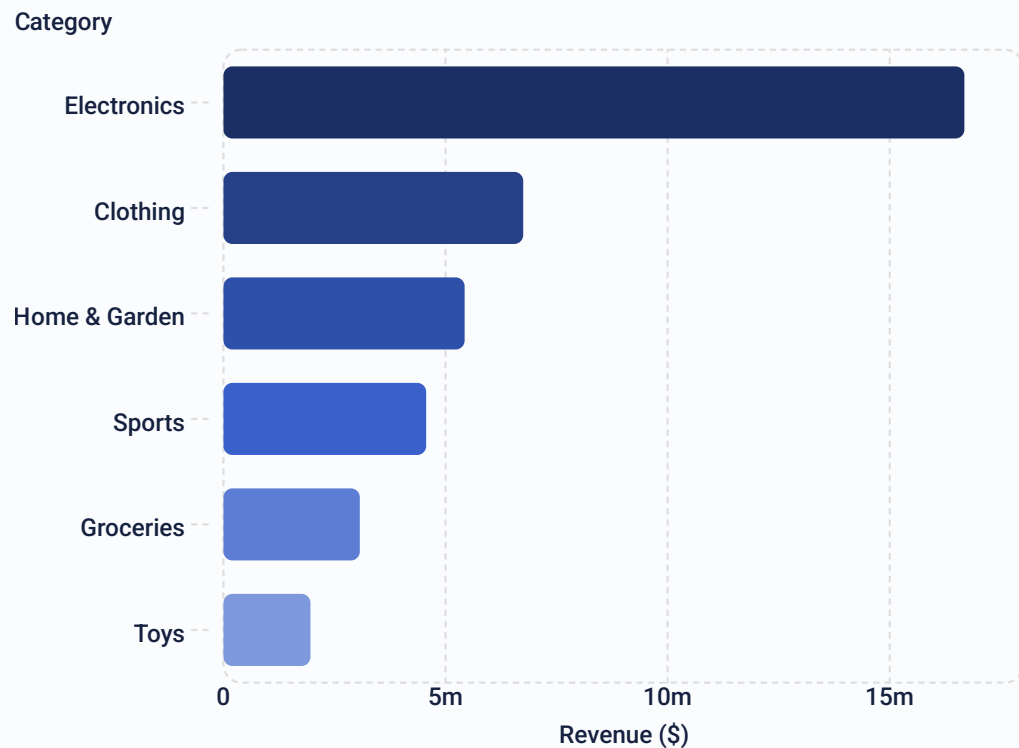
Naive baselines → SARIMA → XGBoost, from simple to sophisticated.

06

Pick a Winner

XGBoost cut forecast error by more than half vs. the naive baseline.

Revenue by Category



Electronics Dominates

Electronics drives 43% of total revenue (\$16.7M) despite moderate unit volume (76,805 units). It's a high-price, high-value category – the one most worth forecasting precisely, since errors there carry the biggest dollar impact.

Groceries lead in unit volume (365,794 units) but contribute only \$3.1M in revenue. Clothing and Home & Garden follow as mid-tier contributors. Total revenue analyzed: **\$38,451,889**.

Key Behavioral Patterns

2.3x

Holiday Revenue Lift

Avg. \$3,881/day vs. \$1,694 on regular days — the strongest single driver in the model.

43%

Promo Unit Lift

53.1 units on promo days vs. 37.2 on non-promo days — a large, measurable effect.

18%

Weekend Lift

Weekend avg. \$1,971 vs. weekday avg. \$1,666 — a reliable, low-noise signal.

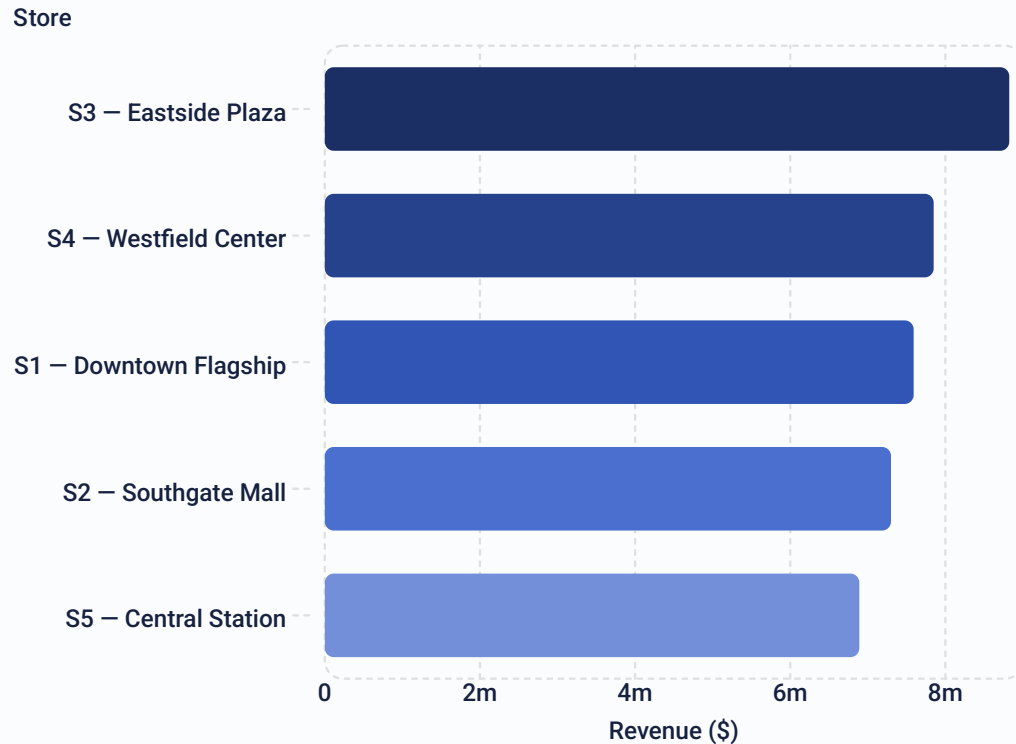
72%

Holiday Feature Weight

`holiday\_flag` accounted for 72% of XGBoost's total feature importance.

❏ These are standing, predictable patterns — not anomalies. They should be built directly into staffing, stocking, and forecasting baselines.

Store Performance



One Shared Model Works

Store performance is **fairly even across all 5 locations**. The range is ~\$1.9M — about 22% of the top store's revenue. No single store is dramatically under- or over-performing.

This supports building **one shared forecasting approach** across stores rather than maintaining five separate models — a meaningful simplification for production deployment.

Forecasting Model Comparison

Evaluated on the last **60 days (Nov 2 – Dec 31, 2024)** — including Black Friday, Cyber Monday, and the Christmas shopping window. A deliberately hard test period.

Model	MAE	RMSE	MAPE
XGBoost (lag + calendar features)	5,285	8,072	6.14%
SARIMA(1,1,1)(1,1,1,7)	10,730	25,451	9.35%
Naive (yesterday repeats)	11,354	21,968	14.15%
7-day Moving Average	16,638	26,674	19.99%
Seasonal Naive (same day last week)	19,999	36,676	23.51%

XGBoost wins because it directly uses `holiday_flag` as an input — pure time-series methods (SARIMA, seasonal naive) must infer holiday spikes from history alone, and both under-predict sharp Black Friday and Christmas peaks. Seasonal naive performs worst because "last week repeats" fails badly when a holiday spike hits.

Why XGBoost Wins

Direct Holiday Input

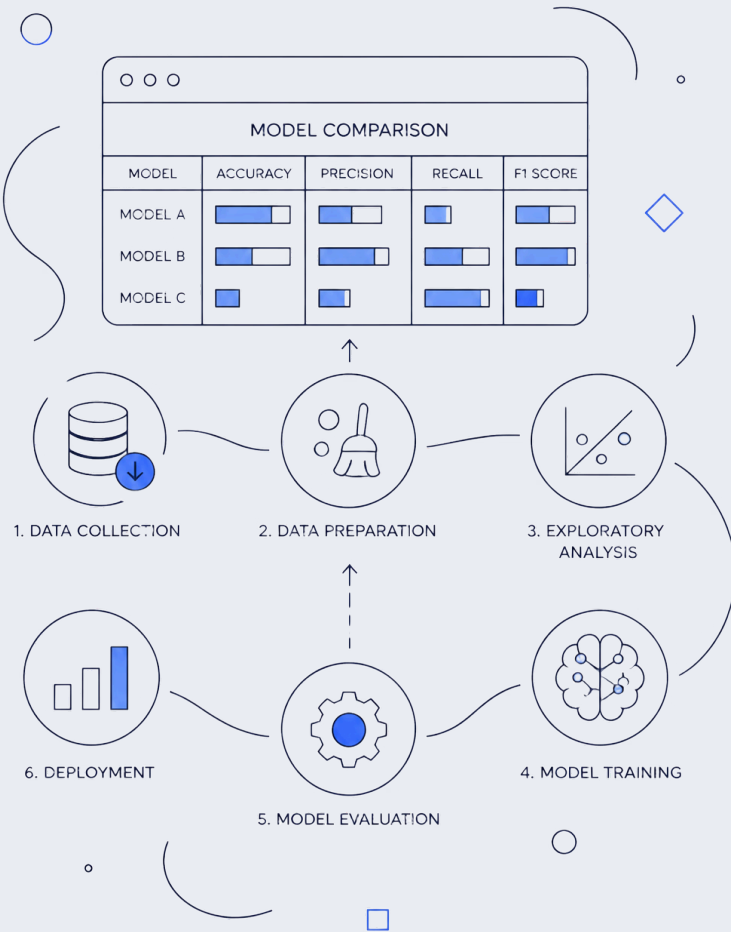
XGBoost takes `holiday_flag` as a direct feature — it knows in advance when a spike is coming. SARIMA and seasonal naive must infer it from historical patterns alone.

Feature Importance: 72%

The holiday flag alone accounted for **72% of the model's predictive weight**, dwarfing lag-based features like yesterday's or last week's revenue.

6.1% MAPE — 2x Better

XGBoost achieved **6.1% average error** vs. 14.2% for the naive baseline — more than a 2x improvement on a genuinely hard test window.



Business Recommendations

1 Prioritize Electronics Forecast Accuracy

At 43% of revenue, a 1% forecasting error in Electronics costs more in absolute dollars than the same error in any other category.

2 Build Holiday & Promo Calendars into Planning

The single highest-leverage input for demand accuracy — inventory planning built around holiday calendars outperforms planning on historical averages alone.

3 Staff & Stock for the ~18% Weekend Lift

Treat this as a standing baseline adjustment, not a special case re-decided each week.

4 Treat Promotions as a ~43% Unit-Volume Lever

Plan promotional calendars against inventory commitments using this measurable lift figure.

5 Use One Shared Model Across All Stores

Store performance is fairly even — a shared forecasting model is a reasonable and simpler starting point.

Project Files & Next Steps

What's in This Project

- `retail_sales_data.csv` – Raw dataset (21,930 rows)
- `retail_sales.db` – SQLite database (`sales`, `dim_store` tables)
- `starter_queries.sql` – 10 ready-to-run EDA queries
- `forecasting_pipeline.py` – Full pipeline: EDA → baselines → SARIMA → XGBoost
- `model_comparison.csv` – MAE/RMSE/MAPE per model
- `Analysis_Report.md` – Findings and recommendations

Quick Start

1. Open `retail_sales.db` in DB Browser for SQLite, run `starter_queries.sql`
2. Read `Analysis_Report.md` for findings
3. Run `forecasting_pipeline.py` to see modeling in action
4. Extend it: per-category forecasts, add Prophet, tune SARIMA

📄 **Limitations & Next Steps:** This report forecasts total daily revenue; a production system would forecast at the **store × category** grain. SARIMA parameters were not exhaustively tuned. Facebook Prophet (explicit holiday regressors) and causal promo-lift methods (difference-in-differences) are worth adding with more data.